Diffusion

Atoms movements in materials

Movement of atoms in **solids**, liquids and gases is very important

Examples: Hardening steel, chrome-plating, gas reactions, Si wafers .. etc.

Today we will discuss:

- Atomic mechanisms of diffusion
- Mathematics of diffusion
- Factors affecting diffusion
- Examples
- Steady state diffusion
- Non-steady state diffusion
- Summary

Diffusion

temp is the vibration of atoms

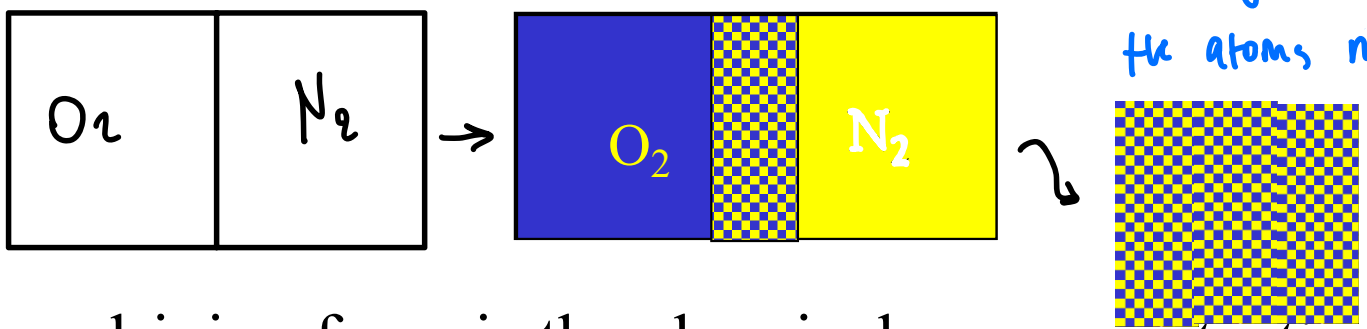
↑ T = ↑ vibration

- Diffusion is the mass transport through atomic motion at high temperature
- can have *self-diffusion* or *interdiffusion* between two materials
  - Self-diffusion occurs in pure elements

ex Fe and Fe

O_2 and N_2

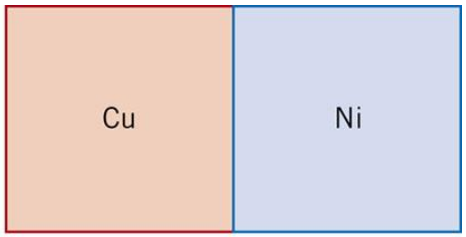
Example: Two chambers, each containing a different gas, separated by a removable barrier; when the barrier is pulled away, *interdiffusion* occurs



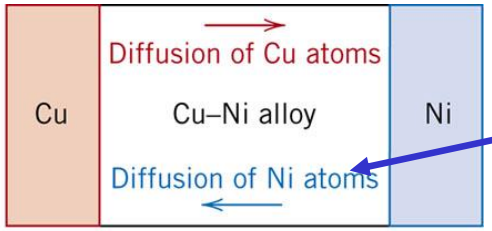
the high temperature makes the atoms move and stick

driving force is the chemical or *concentration gradient* through materials or *diffusion couple*

Diffusion in Solid Materials

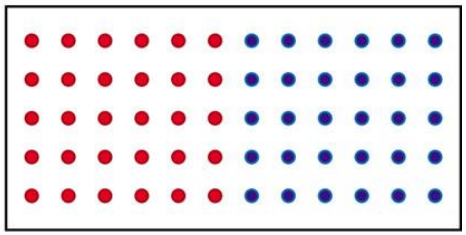


(a)



(a)

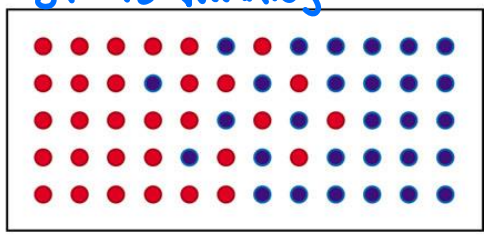
Diffusion Couple



(b)

Heat

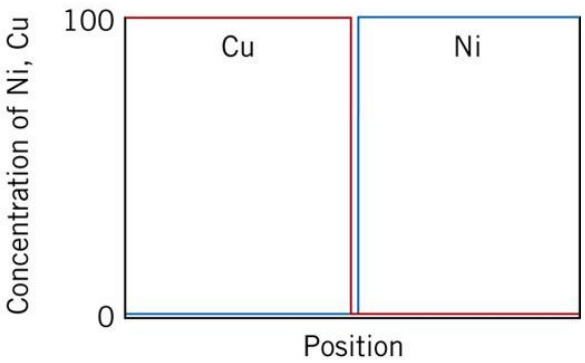
Starts mixing



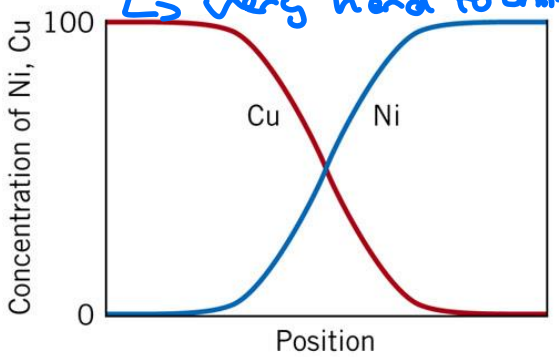
(b)

temperature should be high enough to overcome energy barriers to atomic motion.

permanent mixing
↳ very hard to unmix



(c)



(c)

Heat causes atoms to vibrate

- Vibration amplitude increases with temperature
- If heating continues *melting occurs when vibrations are sufficient to rupture bonds*. Faster diffusion takes place in the liquid state.

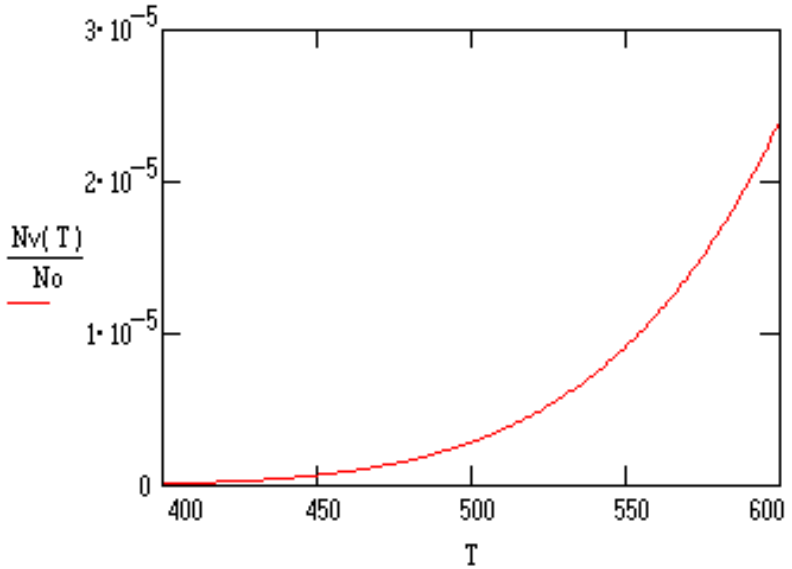
The process of substitutional diffusion requires the presence of ..vacancies..
(Vacancies give the atoms a place to move)

Recall Arhenius equation

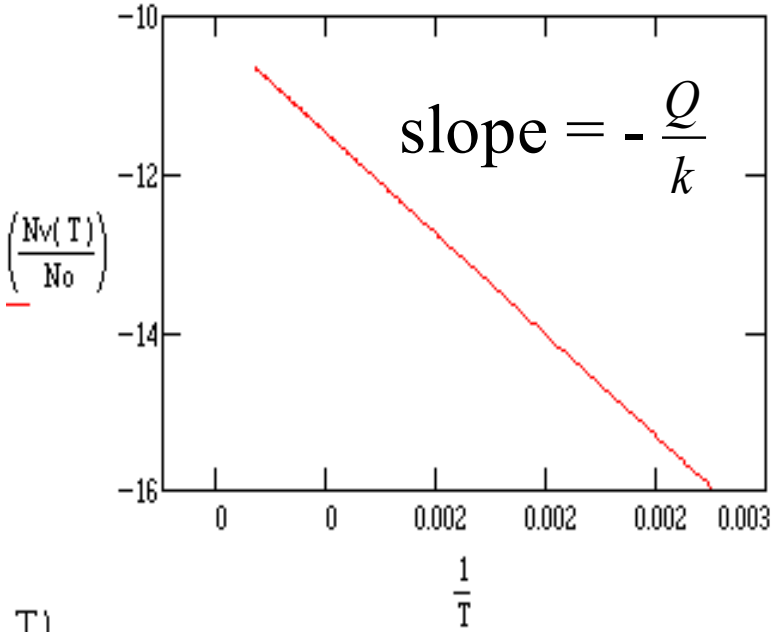
$$N_v = N_o \exp\left(-\frac{Q}{kT}\right)$$

where
N_v = equilibrium # of vacancies
N_o = total number of lattice sites
k = Boltzman's constant
Q = activation energy for vacancy formation

Example: lead (Pb)



Example: lead (Pb)

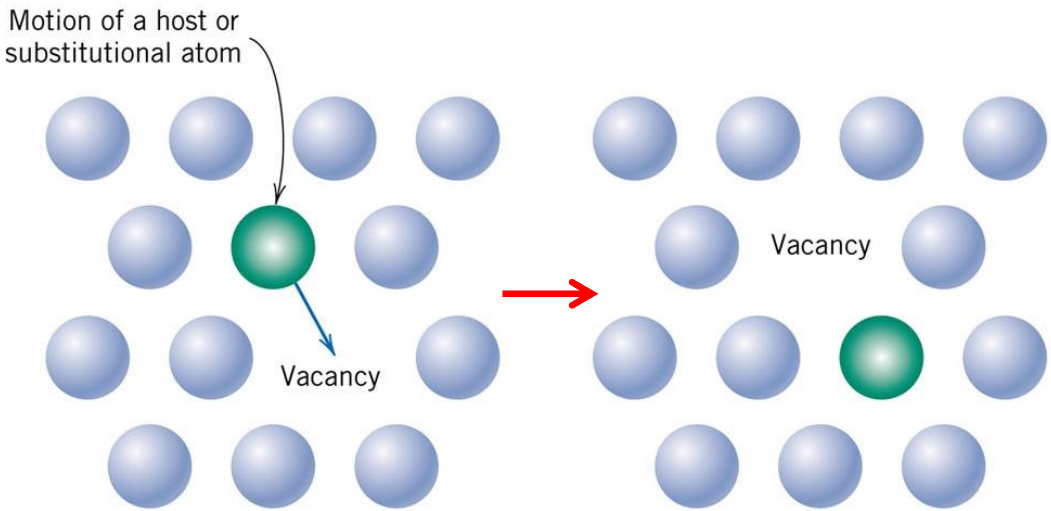


As T ↑, number of vacancies ↑, and energy ↑, so diffusion is faster

↑T → faster diffusion

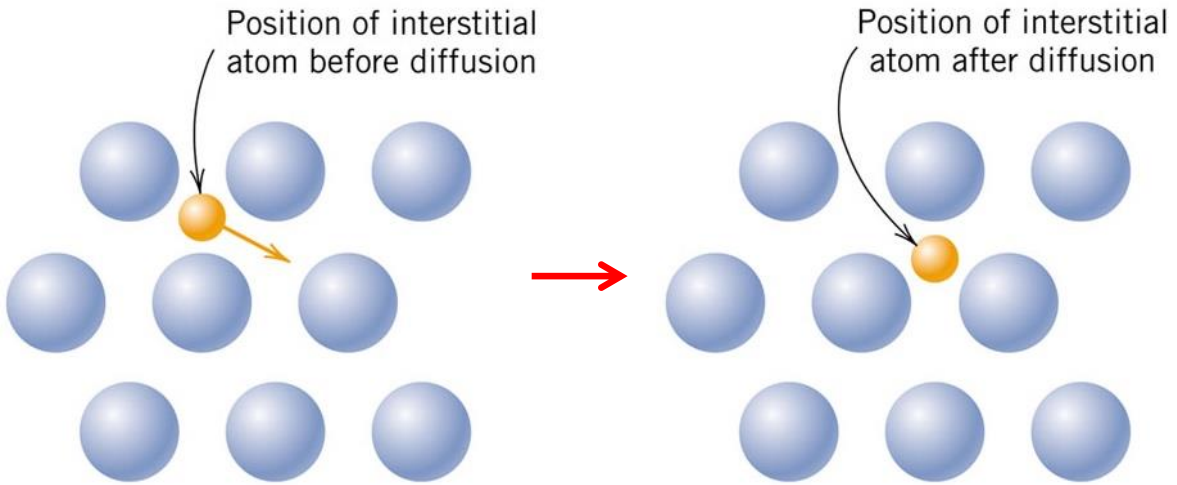
Diffusion Mechanisms

- Atoms are constantly in motion and vibrating
- change of atomic position requires:
  - vacant site
  - energy to break atomic bonds
- Two types of diffusion mechanism:
  - **vacancy** diffusion
  - **interstitial** diffusion
- movement of vacancies in one direction is equivalent to atomic movement in the opposite direction

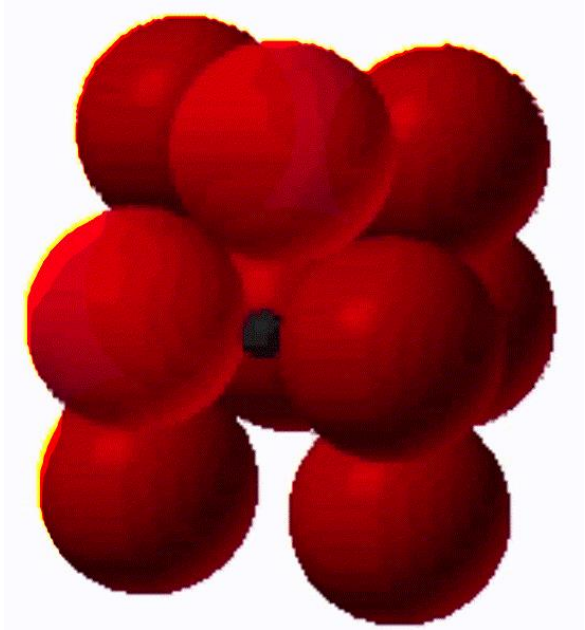


Vacancy diffusion

INTERSTITIAL DIFFUSION



Interstitial diffusion



- Interstitial atom moves from one interstitial site to another (*empty*)
- Energy needed, again to squeeze past atoms.
- Example: …….
- Usually much faster because many more empty interstitial sites and no vacancies are required

Diffusion

- Diffusion is a …time dependant… process

➔ rate of diffusion is important

- diffusion *flux* (J) is defined as the mass, M, diffusing through unit area, A, per unit time, t

A : Area of filter

t = time

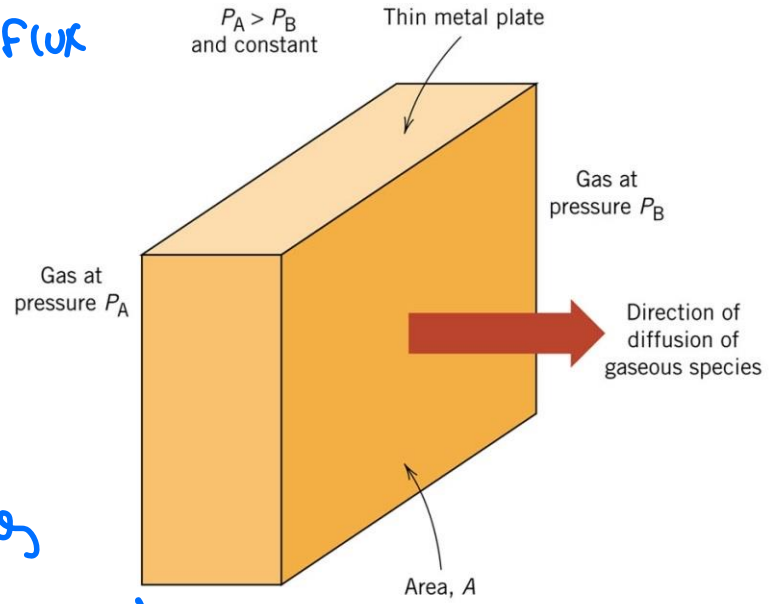
$$J = \frac{M}{At} \quad \text{or} \quad J = \frac{1}{A}\frac{dM}{dt}$$

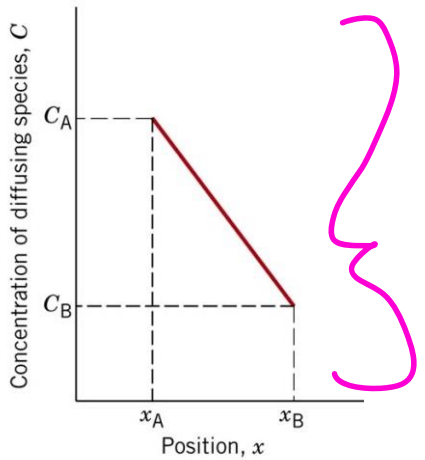
flow inside a "filter"

If flux does not change with time: …steady state……

J = diff flux

↳ constent flux





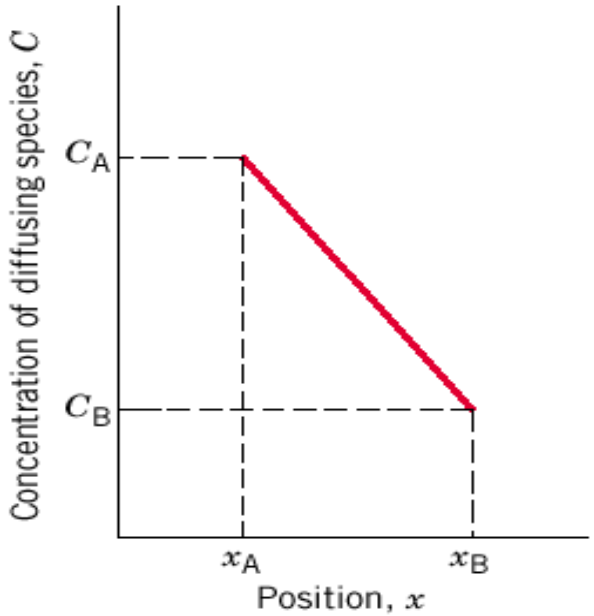
hydrogen diffusing through metal

Steady-State Diffusion

- Concentration gradient = $\frac{dC}{dx}$
- Fick's 1st law:

$$J = -D\frac{dC}{dx}$$

D = -ve



Where D is the ***diffusion coefficient*** (diffusivity) or speed of diffusion *(m^2/s)*.

↳ speed of diffusion

-ve *because atoms diffuse down concentration gradient*

• Example of *steady-state diffusion* is gas diffusing through a metal plate (*gas pressure constant*).

Example

The purification of H_2 (gas) by diffusion through a Pd sheet. Compute the number of kilograms of hydrogen that pass per hour through a 5 mm thick sheet of Pd having an area of 0.20 m^2 at 500°C. Assume a diffusion coefficient of 1.0×10^{-8} m^2/s, that the concentrations at the high and low pressure sides of the plate are 2.4 and 0.6 kg of H_2 per m^3 of Pd, and that steady state conditions have been attained.

Recall, flux is mass per unit time per unit area. Thus, multiplying J by area and time will give total mass.

$$M = JAt = -DAt\frac{\Delta C}{\Delta x}$$

$$= -(1.0\times10^{-8}\ m^2/s)(0.2\ m^2)(3600\ s/h)\left[\frac{0.6-2.4\ kg/m^3}{5\times10^{-3}\ m}\right]$$

$$= 2.6\times10^{-3}\ kg/h$$

interstitial diffusion Case Hardening

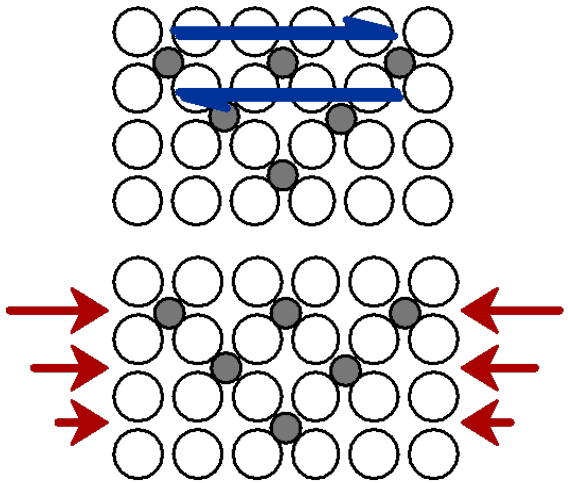


Chapter 5 Callister 8e. (courtesy of Surface Division, Midland-Ross.)

- Example of interstitial diffusion is a case hardened gear.

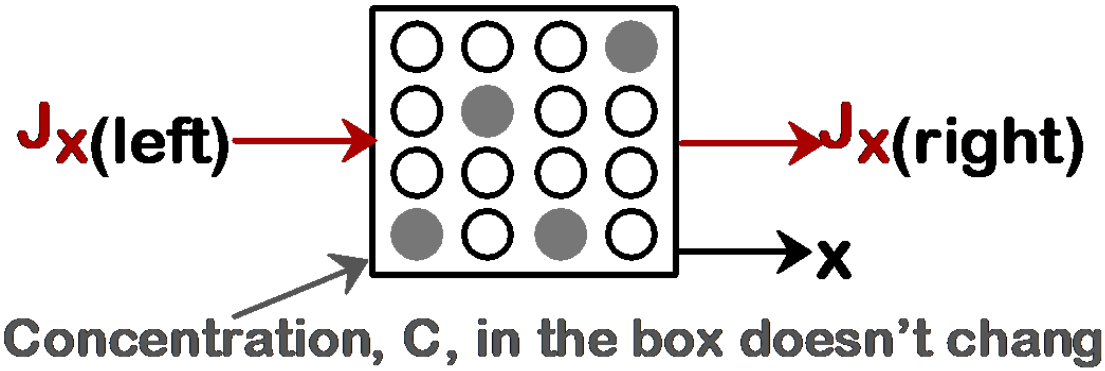
- Diffuse carbon atoms into the host iron atoms at the surface.

- Result: The "Case" is
  - hard to deform: C atoms "lock" planes from shearing.
  - hard to crack: C atoms put the surface in compression.



Steady State Diffusion: *Summary*

- Steady State: the concentration profile doesn't change with time.



Steady State:

$J_x(\text{left}) = J_x(\text{right})$

providing H to left = providing H to right

- **Apply Fick's First Law:** $J_x = -D\frac{dC}{dx}$
- **If $J_{x_left} = J_{x_right}$, then** $\left(\frac{dC}{dx}\right)_{left} = \left(\frac{dC}{dx}\right)_{right}$
- **Result: the slope, dC/dx, must be constant (i.e., slope doesn't vary with position)!**

Non-steady State Diffusion

- In most real situations diffusion is not ……steady state……
- Flux and concentration gradient vary with time

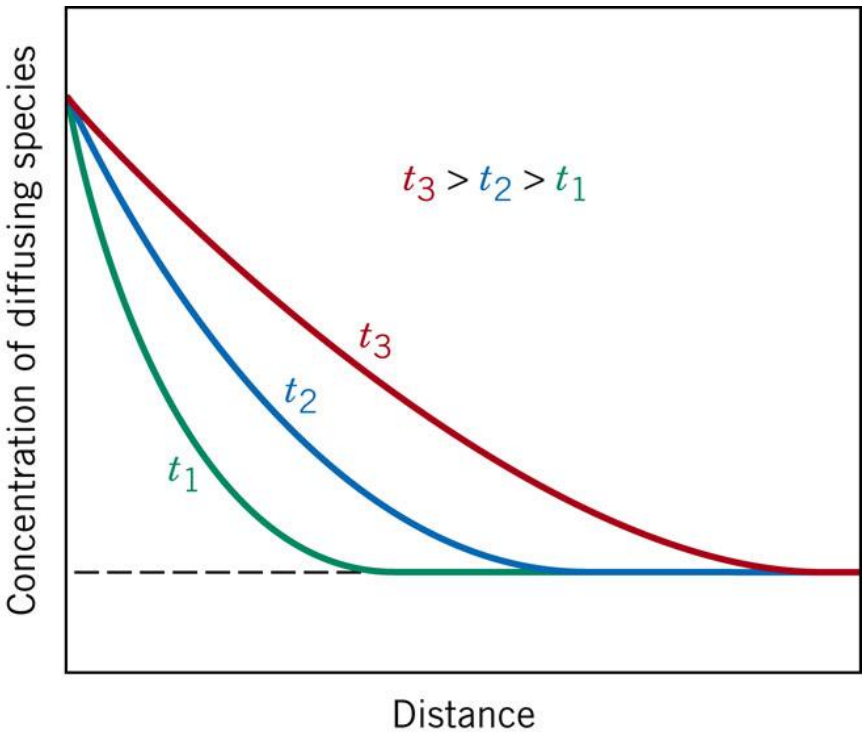


FIGURE 5. 6 Concentration profiles for nonsteady-state diffusion taken at three different times, t_1, t_2, and t_3.

• The changes of the concentration profile is given in this case by a differential equation, ……Fick's…… *second law*: Fick's 2nd law

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right)$$

• If diffusion coefficient is independent of *composition* then: simplified

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Solution of this equation is concentration profile as function of time, ……C(x,t)……



Concentration as a function of time relative to position

Fick's Second Law

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

- Solution requires boundary conditions.
  - *A useful solution is for a semi-infinite solid when the surface concentration remains constant.*
- What is semi-infinite solid?
  - ✓ *semi-infinite bar:* none of the diffusing atoms reaches the bar end during the time over which the diffusion takes place.
  - ✓ $l > 10\sqrt{(Dt)}$, *where* l*: bar length,* D*: diffusion coefficient and* t*: time*

Fick's Second Law - *Application*

- Simple boundary condition is where the surface concentration is held constant,
  - *e.g. gas phase with constant partial pressure at the surface*
- Conditions are:
  - before diffusion, solute atom have a homogeneous concentration of C_o
  - x is zero at the surface and increases with distance into the solid
  - *time is zero just before diffusion begins*

at beg. of membrane

- Mathematically, for $t = 0$, $C = C_o$ at $0 \le x \le \infty$

  $t > 0$, $C = C_s$ at $x = 0$ and $C = C_o$ at $x = \infty$

  C_s = constant surface concentration

- applying these boundary conditions gives:

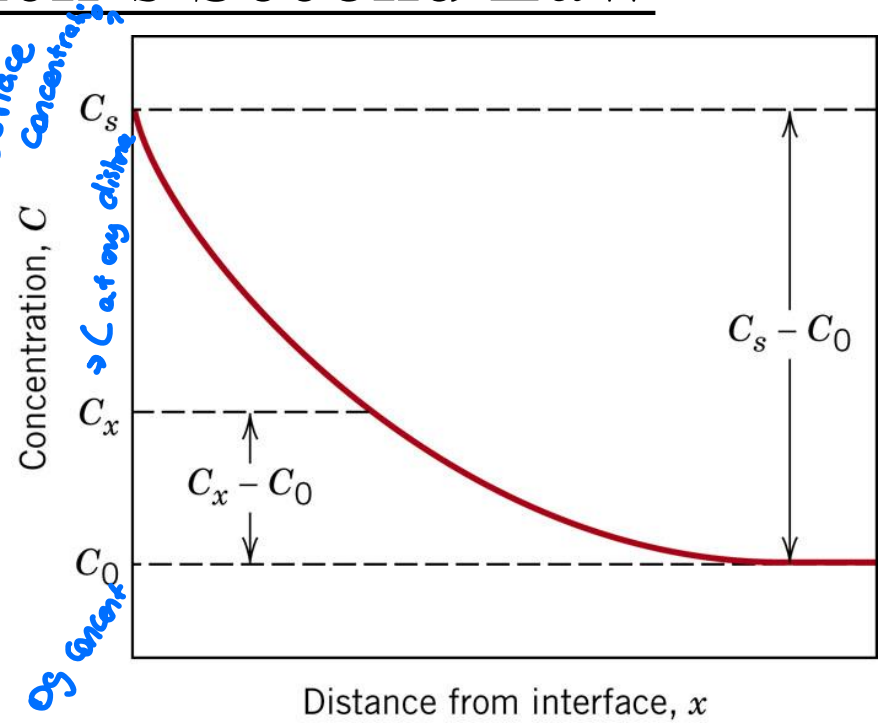
$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

no need to memorize / do ts eq, just apply the variables

error function

Application of Fick's Second Law

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$



C_s = Surface concentration which remains constant

C_0 = Initial concentration in solid

C_x = Concentration at distance x into sample after time t.

D = Diffusivity of solute in solvent, m^2s^{-1}

t = Time, seconds

erf = Gaussian error function, based on integration of the "bell shaped" curve

$$\mathbf{erf}\,(z) = (2/\sqrt{\pi})\int_0^z \exp(-y^2)\,dz \qquad \left(\frac{x}{2\sqrt{Dt}}\right) = z$$

Tabulation of Error Function Values

Table 5.1 Tabulation of Error Function Values

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf(z)$$

$$\left(\frac{x}{2\sqrt{Dt}}\right) = z$$

say $z = 1.1$

then $erf(z) = 0.8802$

in the eq.

Special Case

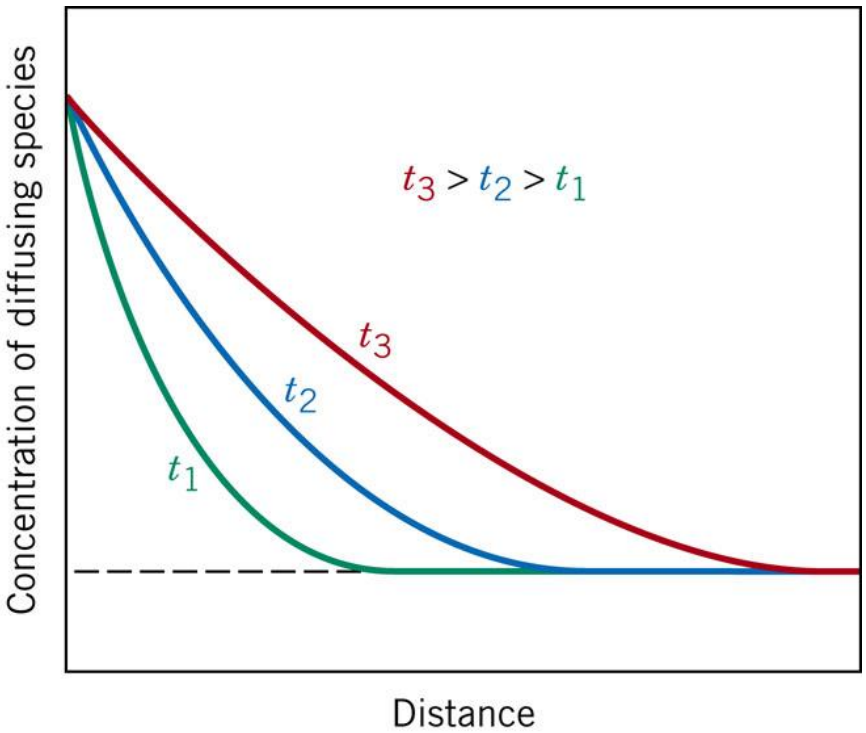


FIGURE 5. 6 Concentration profiles for nonsteady-state diffusion taken at three different times, t_1, t_2, and t_3.

- If it is desired to achieve a particular concentration of solute atoms in an alloy then:

$$\frac{C_1 - C_o}{C_s - C_o} = \text{constant}$$

and

$$\frac{x}{2\sqrt{Dt}} = \text{constant}$$

$$\frac{x^2}{Dt} = \text{constant}$$

- These equations facilitate the diffusion computation for this specific case (*see example 5.2*).

Factors Affecting Solid-State Diffusion

- Diffusing species and host material are important
  - smaller **atoms** can "squeeze" in between host atoms more easily
  - in lower **packing density** host material ⇒ easier for atoms to migrate with fewer bonds to expand
    - eg faster in more open lattice (BCC faster than FCC) *(APF....)*
  - in lower **melting point** host material ⇒ weaker bonds (easier to push apart)

Smaller atom diffuse easier

↑MP = harder diffusion

- *Temperature* has a strong effect on diffusion rates:

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

Where: D_o determines if its self diffusion or not

D_o = temperature dependant pre-exponential constant

Q_d = *activation energy* for diffusion

R = gas constant (8.31 J/mol K)

T = absolute temperature

Activation energy is the barrier to diffusion and controls the diffusion coefficient, D

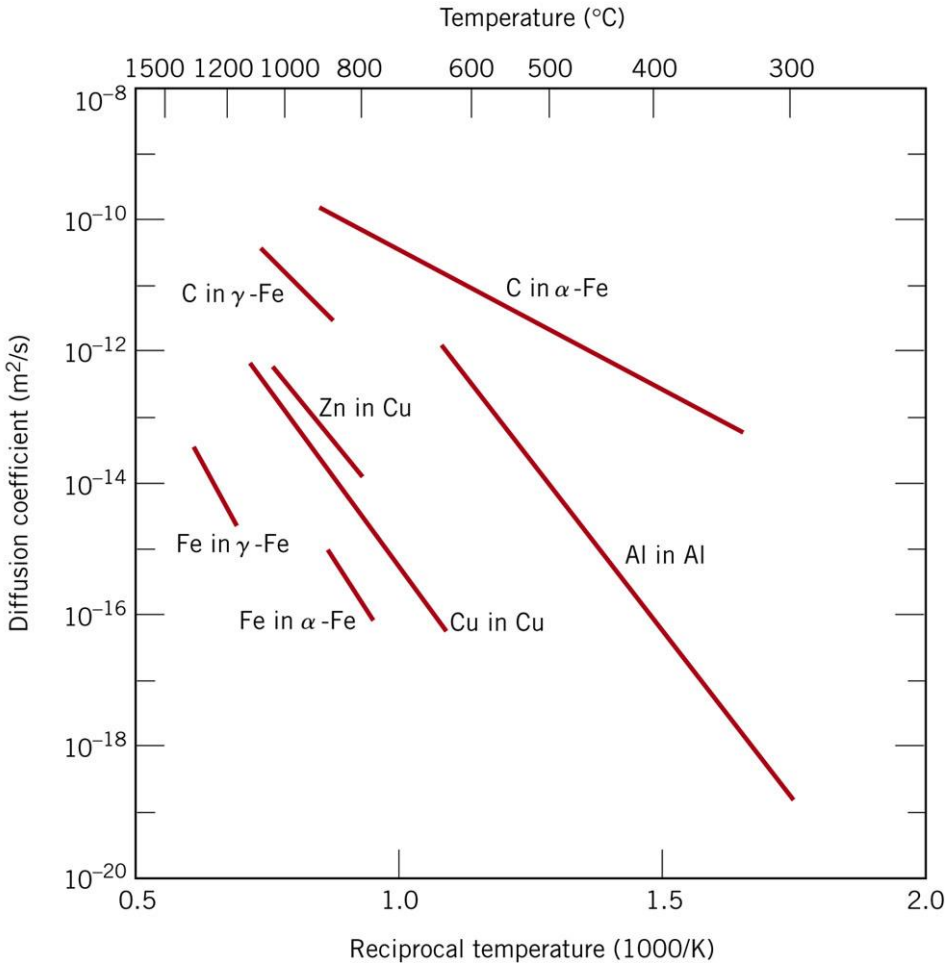
Typical Diffusion Data

Host crystal	Atom	D_o (cm^2/s)	q_d (eV/atom)
Cu	Cu	0.20	2.04
Cu	Zn	0.34	1.98
Ag	Ag	0.40	1.91
Ag	Au	0.26	1.98
Ag	Cu	1.2	2.00
Ag	Pb	0.22	1.65
U	U	0.002	1.20
Si	Al	8.0	3.47
Si	Ga	3.6	3.51
Si	In	16.0	3.90

Typical values for preexponential (D_o) and activation energy

(after Kittel, "Solid State Physics" 5th ed.)

Experimental Determination of Activation Energy



$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

- Taking logs:

$$\log D = \log Do - \frac{Qd}{2.3R}\left[\frac{1}{T}\right]$$

- plot log D vs. 1/T
- slope = $-Q_d/2.3R$
- intercept = D_o

FIGURE 5.6 Plot of the logarithm of the diffusion coefficient versus the reciprocal of absolute temperature for several metals.

Example

0,35% → 0% C decarburization C – 0

An FCC Fe-C alloy initially containing 0.35 wt. % C is exposed to an oxygen-rich (and carbon-free) atmosphere at 1400 K (1127°C). Under these conditions the carbon in the alloy diffuses toward the surface and reacts with the oxygen in the atmosphere; that is, the carbon concentration at the surface is maintained essentially at 0 wt. % C. (This process of carbon depletion is termed decarburization). At what position will the carbon concentration be 0.15 wt. % after a 10-hour treatment. The value of D at 1400 K is 6.9×10^{-11} m^2 /s.

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0,15 - 0,35}{0 - 0,35} = 0,5714 = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 0,4284 \qquad z = 0,4$$

$$x = 2(0,4)\sqrt{Dt} = (0,8004)\sqrt{(6,9 \times 10^{-11}\ m^2/s)(3,6 \times 10^4\ s)}$$

$$1,3 \times 10^{-3}\ m = 1,3\ mm$$

SUMMARY

Is diffusion faster, slower or same in the following scenarios?

1- Close-packed vs open structure

slower ✓ faster

2- Lower-melting vs higher-melting-point materials

faster ✓ slower

3- Covalent bond vs secondary interactions

slower ✓ faster

4- Smaller vs larger atoms

faster ✓ slower

Use the QR code shown in the classroom. This one won't work.

Next topic:

Mechanical properties of materials